We use $\mathbb{Z}$ to denote the set of all **Integers**.

$$\mathbb{Z} = \{ \ldots, -3, -2, -1, 0, 1, 2, 3, \ldots \}.$$

When you look at the figures (Fig. 2.2 and 2.3) above, you are sure to get amused by the gap between any pair of consecutive integers. Could there be some numbers in between?

You have come across fractions already. How will you mark the point that shows $\frac{1}{2}$ on $\mathbb{Z}$? It is just midway between 0 and 1. In the same way, you can plot $\frac{1}{3}, \frac{1}{4}, \frac{1}{5}, 2\frac{3}{4}$ ... etc. These are all fractions of the form $\frac{a}{b}$ where a and b are integers with one restriction that $b \neq 0$. (Why?) If a fraction is in decimal form, even then the setting is same.

Because of the connection between fractions and ratios of lengths, we name them as **Rational numbers**. Here is a rough picture of the situation:



Fig. 2.4

A **rational number** is a fraction indicating the quotient of two integers, excluding division by zero.

Since a fraction can have many equivalent fractions , there are many possible forms for the same rational number. Thus $\frac{1}{3}, \frac{2}{6}, \frac{8}{24}$ all these denote the same rational number.

2.2.1 Denseness Property of Rational Numbers

Consider a, b where $a > b$ and their AM(Arithmetic Mean) given by $\frac{a+b}{2}$. Is this AM a rational number? Let us see.

If $a = \frac{p}{q}$ (p, q integers and $q \neq 0$); $b = \frac{r}{s}$ (r, s integers and $s \neq 0$), then

$$\frac{a+b}{2} = \frac{\frac{p}{q} + \frac{r}{s}}{2} = \frac{ps+qr}{2qs}$$ which is a rational number.

We have to show that this rational number lies between a and b.

$a - \left(\frac{a+b}{2}\right) = \frac{2a-a-b}{2} = \frac{a-b}{2}$ which is > 0 since $a > b$.

Therefore, $a > \left(\frac{a+b}{2}\right)$ ... (1)

$\left(\frac{a+b}{2}\right) - b = \frac{a+b-2b}{2} = \frac{a-b}{2}$ which is > 0 since $a > b$.

Therefore, $\left(\frac{a+b}{2}\right) > b$ ... (2)

From (1) and (2) we see that $a > \left(\frac{a+b}{2}\right) > b$, which can be visualized as follows:

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